



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

gRPC PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A API Technologies

TOTAL: 10 QUESTIONS

Q1 What is gRPC and what API communication problem does it solve?

Threat Model

Q6 How do you implement pagination in a gRPC API?

Observability

Q2 How does gRPC define its schema or contract?

Architecture

Q7 How do you document a gRPC API?

Lifecycle

Q3 How does gRPC handle authentication?

Configuration

Q8 How do you test a gRPC API?

Compliance

Q4 How does gRPC handle API versioning?

Hardening

Q9 How do you protect a gRPC API from abuse?

Testing

Q5 How does gRPC handle errors and status codes?

SSO / IdP

Q10 What monitoring and observability should a gRPC API have?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 gRPC defines a protocol or standard

Mitigation

gRPC defines a protocol or standard for structured communication between services. It addresses concerns like schema definition, payload format, versioning, or transport efficiency that plain HTTP with ad-hoc JSON does not solve on its own.

A6 Cursor-based pagination (next_cursor token) is

Observability

Cursor-based pagination (next_cursor token) is preferred for large or frequently-updated datasets because it is stable. Offset-based pagination (page, limit) is simpler but can skip or duplicate rows if data changes between requests.

A2 gRPC uses an IDL or schema

Primitives

gRPC uses an IDL or schema language to declare types, operations, and their inputs and outputs. This schema is the single source of truth from which documentation, validation, and client SDKs can be auto-generated.

A7 gRPC APIs use an API specification

Automation

gRPC APIs use an API specification (OpenAPI, AsyncAPI, GraphQL SDL) to generate interactive documentation (Swagger UI, ReDoc, GraphiQL). Good docs include examples, error codes, and authentication instructions.

A3 gRPC APIs commonly use Bearer tokens

Hardening

gRPC APIs commonly use Bearer tokens (JWT/OAuth2) in the Authorization header or API keys in custom headers. Transport-level security (TLS) is always required. Additional mechanisms (mTLS, HMAC) apply to high-trust scenarios.

A8 Contract testing (Pact) verifies the provider

Governance

Contract testing (Pact) verifies the provider matches the consumer's schema. Integration tests call real endpoints against a test environment. Load tests (k6, Gatling) verify performance under production-like concurrency.

A4 Common versioning strategies include URL path

Gotchas

Common versioning strategies include URL path (/v1/users), request headers (Accept: application/vnd.api+v2+json), and query params (?version=2). gRPC prefers one strategy and maintains backward compatibility within a major version.

A9 Rate limiting (tokens per IP per

Assessment

Rate limiting (tokens per IP per minute) prevents DoS. Input validation rejects malformed requests before they reach business logic. Output filtering (response schema) prevents accidental data leakage. Monitor for anomalous usage patterns.

A5 gRPC uses HTTP status codes (200

Federation

gRPC uses HTTP status codes (200 OK, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Internal Server Error) combined with a structured error body containing a code, message, and optional details field.

A10 Instrument request count, latency histograms,

Optimization

Instrument request count, latency histograms, and error rate with Prometheus or a cloud metrics system. Add distributed tracing with OpenTelemetry. Log structured request/response data. Set SLOs and alert when SLIs breach them.